

Hodge-dual hyperplanes give an infinite family of Plücker ETFs

Candidate full answer to Question 6.3 of arXiv:1703.01786

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Abstract

Question 6.3 asks whether Plücker equiangular tight frames (ETFs) exist with parameters different from the paper's single $(16, 6)$ example, lifted from a tight $(16, 2, 4)$ fusion frame. We prove a transfer principle: every real unit-norm (n, t) ETF, for $t \geq 3$, is the lifted Plücker image of the tight rank- $(t-1)$ hyperplane fusion frame $Q_j = I - u_j u_j^T$. The reason is that Hodge star is an isometry $\mathbb{R}^t \rightarrow \bigwedge^{t-1} \mathbb{R}^t$ and identifies a normal vector with the Plücker vector of its orthogonal hyperplane. Applying this to regular simplices gives, for every $t \geq 3$, a Plücker ETF with parameters $(n, m, l, t) = (t+1, t, t-1, t)$. Thus the question has an affirmative answer by an infinite family, already beginning with a $(4, 3)$ ETF lifted from a tight $(4, 2, 3)$ fusion frame.

1 The source question

The source defines an equiangular (n, m) -frame to be a *Plücker ETF* when it is the lifted Plücker image of a tight (n, l, t) fusion frame, where $l \geq 2$ and $m = \binom{t}{l}$. After constructing one example with $(n, m, l, t) = (16, 6, 2, 4)$, it asks whether examples with different parameters exist.

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using the H_j s as building blocks. Noting that $W_{4,3} = \frac{1}{3}$, we conclude that F is the synthesis matrix of a Steiner ETF, $\mathcal{F}$, for $\mathbb{R}^6$ consisting of 16 vectors. In light of this construction, we conclude with a natural question, the posing of which requires a definition.

DEFINITION 6.2. An equiangular (n, m) -frame for $\mathbb{F}^m$, $\mathcal{H}$, with synthesis matrix, $H = [h_1 h_2 \dots h_n]$, is a **Plücker ETF** if there exists a (real or complex) tight (n, l, t) -fusion frame $\mathcal{Q} = \{Q_j\}_{j=1}^n$, where $l \geq 2$, $m = \rho_{t,l}$ and $f_j = \overline{\Phi}(Q_j)$ for every $j \in \{1, 2, \dots, n\}$.

Thus, the Steiner ETF, $\mathcal{F}$, that we have just constructed is a Plücker ETF. Given the volumetric nature of the Plücker embedding (ie, the determinants of the submatrices correspond to (signed) hyper-volumes of hyper-parallopipeds) and that - as an astute reader might have noticed - the generators for each P_j from our example are essentially 2-simplices living in $\mathbb{R}^4$, we find our construction strangely intuitive. Nevertheless, we are unaware of any other examples.

QUESTION 6.3. Other than the Plücker ETF, $\mathcal{F}$, constructed in this section, do Plücker ETFs with different parameters exist?

The answer below uses $l = t - 1$. This is not the degenerate orthonormal-basis case: regular simplices have $n = t + 1 > t = m$ and nonzero common angle $1/t$.

2 The hyperplane transfer theorem

We use the Euclidean Hodge isometry

$$* : \mathbb{R}^t \longrightarrow \bigwedge^{t-1} \mathbb{R}^t.$$

After fixing the standard ordered basis, the coordinates of a decomposable vector $a_1 \wedge \cdots \wedge a_{t-1}$ are precisely the maximal minors of the matrix whose rows are $a_1^\top, \dots, a_{t-1}^\top$. They are therefore precisely its Plücker coordinates.

Theorem 1 (ETF-to-hyperplane Plücker transfer). *Let $t \geq 3$, and let $\mathcal{H} = \{u_j\}_{j=1}^n \subset \mathbb{R}^t$ be a real unit-norm (n, t) ETF. Put*

$$Q_j = I_t - u_j u_j^\top \quad (1 \leq j \leq n).$$

Then $\mathcal{Q} = \{Q_j\}_{j=1}^n$ is a tight $(n, t-1, t)$ fusion frame, and its lifted Plücker vectors form an (n, t) ETF. Consequently that ETF is a Plücker ETF in the sense of the source paper. Moreover, $\mathcal{Q}$ is itself chordally equiangular.

Proof. Since u_j is a unit vector, Q_j is the orthogonal projection onto $u_j^\perp$ and has rank $t-1$. Tightness of $\mathcal{H}$ says

$$\sum_{j=1}^n u_j u_j^\top = \frac{n}{t} I_t.$$

Hence

$$\sum_{j=1}^n Q_j = n I_t - \frac{n}{t} I_t = \frac{n(t-1)}{t} I_t,$$

which is exactly the required tight fusion-frame identity.

For each j , choose an orthonormal basis $a_{j,1}, \dots, a_{j,t-1}$ of $u_j^\perp$, and let A_j be the $(t-1) \times t$ matrix having the $a_{j,r}^\top$ as rows. Then $A_j A_j^\top = I_{t-1}$ and $A_j^\top A_j = Q_j$, so A_j is a generator of Q_j in the terminology of the source. Its Plücker vector is the coordinate vector of

$$a_{j,1} \wedge \cdots \wedge a_{j,t-1}.$$

This is a unit generator of the one-dimensional space $\bigwedge^{t-1}(u_j^\perp)$. Therefore, for some sign $\varepsilon_j \in \{\pm 1\}$,

$$\overline{\Phi}(Q_j) = \varepsilon_j * u_j. \quad (1)$$

The sign records the selected orientation class of the generator. A different allowed choice only changes ε_j , which has no effect on tightness or on absolute inner products.

Hodge star is an isometry, and multiplication of individual frame vectors by signs preserves the frame operator. Thus (1) gives

$$|\langle \overline{\Phi}(Q_j), \overline{\Phi}(Q_k) \rangle| = |\langle u_j, u_k \rangle| \quad (j \neq k)$$

and

$$\sum_{j=1}^n \overline{\Phi}(Q_j) \overline{\Phi}(Q_j)^\top = \frac{n}{t} I_{\binom{t}{t-1}} = \frac{n}{t} I_t.$$

The lifted vectors are consequently a unit-norm (n, t) ETF. Since $t-1 \geq 2$ and $\binom{t}{t-1} = t$, this is a Plücker ETF under the source definition.

Finally, if the common ETF coherence is c , then for $j \neq k$,

$$\text{tr}(Q_j Q_k) = \text{tr}((I - u_j u_j^\top)(I - u_k u_k^\top)) = t - 2 + |\langle u_j, u_k \rangle|^2 = t - 2 + c^2.$$

Thus the fusion frame is chordally equiangular as well. $\square$

3 An infinite family with new parameters

Corollary 2 (Regular-simplex family). *For every integer $t \geq 3$, there exists a real Plücker ETF with*

$$(n, m, l, t) = (t + 1, t, t - 1, t).$$

In particular these give infinitely many parameter sets different from $(16, 6, 2, 4)$.

Proof. Let $V = \mathbf{1}^\perp \subset \mathbb{R}^{t+1}$, identified isometrically with $\mathbb{R}^t$, and set

$$u_j = \sqrt{\frac{t+1}{t}} \left(e_j - \frac{1}{t+1} \mathbf{1} \right), \quad 1 \leq j \leq t+1.$$

These vectors have norm one, satisfy $\langle u_j, u_k \rangle = -1/t$ for $j \neq k$, and obey

$$\sum_{j=1}^{t+1} u_j u_j^\top = \frac{t+1}{t} I_V.$$

They are the regular-simplex $(t+1, t)$ ETF. Theorem 1 now applies, with $l = t - 1 \geq 2$ and $m = \binom{t}{t-1} = t$. $\square$

The first member: a tetrahedral $(4, 3)$ Plücker ETF

In $\mathbb{R}^3$, take

$$\begin{aligned} u_1 &= 3^{-1/2}(1, 1, 1), & u_2 &= 3^{-1/2}(1, -1, -1), \\ u_3 &= 3^{-1/2}(-1, 1, -1), & u_4 &= 3^{-1/2}(-1, -1, 1). \end{aligned}$$

Every off-diagonal inner product is $-1/3$. The four rank-two projections $Q_j = I_3 - u_j u_j^\top$ satisfy

$$\sum_{j=1}^4 Q_j = \frac{8}{3} I_3.$$

For example, an oriented orthonormal generator of $u_1^\perp$ has rows

$$a = \frac{1}{\sqrt{2}}(1, -1, 0), \quad b = \frac{1}{\sqrt{6}}(1, 1, -2).$$

In the Plücker coordinate order $(12, 13, 23)$, its three maximal minors are

$$(\det[a_1 \ a_2; b_1 \ b_2], \det[a_1 \ a_3; b_1 \ b_3], \det[a_2 \ a_3; b_2 \ b_3]) = \frac{1}{\sqrt{3}}(1, -1, 1).$$

This is $*u_1$ under the standard coordinate identification $*(x_1, x_2, x_3) = (x_3, -x_2, x_1)$. The other three generators give $\pm *u_j$, so their Plücker images are exactly a tetrahedral $(4, 3)$ ETF. This is lifted from a tight $(4, 2, 3)$ fusion frame and is already a concrete parameter set distinct from the source example.

4 Checks, scope, and novelty guardrail

The proof uses only two identities: complementation converts the frame operator into the tight fusion-frame operator, and Hodge star identifies the normal line with the top exterior power of its hyperplane. The determinant coordinates in the source definition are exactly the coordinates of that exterior product. Orientation ambiguity produces only independent signs.

The result answers Question 6.3 as literally posed, and more strongly gives an infinite family with $n > m$ and nonzero common angle. It does not claim that these examples reproduce the source’s chordally *biangular* $(16, 2, 4)$ precursor; rather, their precursors are chordally equiangular, which is permitted by Definition 6.2 and follows from the final calculation in Theorem 1.

On 11 August 2026, bounded searches of the run indexes and arXiv-facing web results were made for the exact arXiv identifier, “Plücker ETF,” “Plucker ETF,” “Plücker embedding equiangular tight frame,” and Hodge star/hyperplane variants. They recovered the source paper and general ETF or fusion-frame literature, but no later answer or this transfer construction. This is a novelty guardrail, not a comprehensive priority claim.

Review focus. The only convention-sensitive point is (1): a reader should check the source’s chosen ordering of maximal minors against the chosen Hodge-star coordinate ordering. Any discrepancy is a fixed signed permutation, hence an ambient orthogonal map, and any generator-orientation discrepancy is an individual sign. Neither changes the ETF conclusion.